

Skills Summary

Writing Growth and Decay Models from Data

Use this reference while completing your worksheet or when studying.

Skill 1 — The multiplier from a table

Model shape: $y = a \cdot r^t$, where a is the amount at $t = 0$ and r is what you multiply by in each period.

From a table over equal time steps: r is the ratio of consecutive values. Check it on a second pair — if the ratio changes, the data is not exponential.

Reading r : $r > 1$ is growth, $0 < r < 1$ is decay, and r is never negative.

Example: Values 200, 240, 288 at $t = 0, 1, 2$. Find r and predict $t = 3$.

Step 1. Equal time steps means the ratio test is the right tool, and the value at $t = 0$ is already the starting amount a .

Step 2. $240 \div 200 = 1.2$, and $288 \div 240 = 1.2$ too, so the model is $200 \cdot 1.2^t$.

Step 3. At $t = 3$: $1.2^3 = 1.728$, so 200×1.728 .

$r = 1.2$ and value at $t = 3$ is **345.6**

Try it: Values 500, 450, 405 at $t = 0, 1, 2$. Find r and the value at $t = 3$.

check: $r = 0.9$, value = **364.5**

Skill 2 — The multiplier from a percent

Growth of p percent: $r = 1 + \frac{p}{100}$ — you keep the whole and add p hundredths. So 6% growth is $r = 1.06$.

Decay of p percent: $r = 1 - \frac{p}{100}$ — you keep what is left. So 20% decay is $r = 0.8$.

The commonest error: taking the percent of the *original* amount every period. Each period's change is a percent of the amount at the *start of that period*, which is what raising r to a power does for you.

Example: \$1500 grows 6% a year. What is it worth after 5 years?

Step 1. A fixed percent per period is exponential, so convert the percent to a multiplier and raise it to the number of periods.

Step 2. $r = 1 + \frac{6}{100} = 1.06$, so the model is $1500 \cdot 1.06^t$.

Step 3. $1.06^5 = 1.3382255776$, so $1500 \times 1.3382255776 = 2007.338 \dots$

$r = 1.06$, value = **2007.34** dollars

Try it: A \$900 machine loses 20% of its value a year. What is it worth after 4 years?

check: $r = 0.8$, value = **498.96**

Skill 3 — Solving with the model

Doubling and half-life models: “doubles every k periods” is $a \cdot 2^{t/k}$; “halves every k periods” is $a \cdot \left(\frac{1}{2}\right)^{t/k}$. The t/k counts how many doublings or halvings have happened.

To solve for t : divide by a first, write the result as a power of the same base, then match exponents and solve the small equation in t .

Example: A count follows $80 \cdot 2^{t/4}$. When does it reach 640?

Step 1. The unknown is in the exponent, so clear the coefficient and rewrite both sides as powers of two — no logarithm needed.

Step 2. Divide by 80: $2^{t/4} = 8 = 2^3$, so $\frac{t}{4} = 3$.

Step 3. Multiply by 4. Check: $80 \cdot 2^3 = 640$.

$$t = 12$$

Try it: A mass follows $96 \cdot \left(\frac{1}{2}\right)^{t/6}$. When does it reach 12?

check: $t = 18$